6	(a)		Into the page / into the (plane of) paper. (Do not accept into the plane unless candidate specify which plane they are referring to)	
	(b)		Magnetic force (acting on ions) provides the centripetal force (or centripetal acceleraton) $Bqv = \frac{mv^2}{r}$ or $Bq(v \sin\theta) = \frac{m(v \sin\theta)^2}{r}$ $B = \frac{(20\times1.66\times10^{-27})(1.40\times10^{5})}{(1.60\times10^{-19})\left(\frac{0.128}{2}\right)}$ $= 0.4539 \approx 0.454$ T (shown)	
	(c)	(i)	Proper semi-circle with **<u>slightly larger</u>** diameter	

		(ii)	From $Bqv = \frac{mv^2}{r}$ $\quad\quad B \propto m$ $\frac{B}{0.454} = \frac{22}{20}$ $B = 0.499$ T Or $\quad B = \frac{(22\times1.66\times10^{-27})(1.40\times10^{5})}{(1.60\times10^{-19})\left(\frac{0.128}{2}\right)}$ $\quad B = 0.499$ T	

7	(a)			